Advanced decision-making using patient-reported outcome measures in total joint replacement

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Journal:	Publication Year:	Citations:	DOI:
Journal of Orthopaedic Research®	2020.0	53	10.1002/jor.24614

CLINICAL & AGENTIC AI TAKEAWAY

A shared decision-making tool using patient-reported outcomes and clinical data can personalize total joint replacement risks and benefits, improving decision quality and patient satisfaction.

ABSTRACT

Up to one-third of total joint replacement (TJR) procedures may be performed inappropriately in a subset of patients who remain dissatisfied with their outcomes, stressing the importance of shared decision-making. Patient-reported outcome measures capture physical, emotional, and social aspects of health and wellbeing from the patient's perspective. Powerful computer systems capable of performing highly sophisticated analysis using different types of data, including patient-derived data, such as patient-reported outcomes, may eliminate guess work, generating impactful metrics to better inform the decision-making process. We have created a shared decision-making tool which generates personalized predictions of risks and benefits from TJR based on patient-reported outcomes as well as clinical and demographic data. We present the protocol for a randomized controlled trial designed to assess the impact of this tool on decision quality, level of shared decision-making, and patient and process outcomes. We also discuss current concepts in this field and highlight opportunities leveraging patient-reported data and artificial intelligence for decision support across the care continuum. © 2020 Orthopaedic Research Society. Published by Wiley Periodicals, Inc.

Keywords: Artificial intelligence, Arthroplasty, Shared Decision Making, Proms, Hip/knee Osteoarthritis

Study Overview & Background

Up to one-third of total joint replacement (TJR) procedures may be performed inappropriately in a subset of patients who remain dissatisfied with their outcomes, stressing the importance of shared decision-making. Patient-reported outcome measures capture physical, emotional, and social aspects of health and wellbeing from the patient's perspective. Powerful computer systems capable of performing highly sophisticated analysis using different types of data, including patient-derived data, such as patient-reported outcomes, may eliminate guess work, generating impactful metrics to better inform the decision-making process. We have created a shared decision-making tool which generates personalized predictions of risks and benefits from TJR based on patient-reported outcomes as well as clinical and demographic data. We present the protocol for a randomized controlled trial designed to assess the impact of this tool on decision quality, level of shared decision-making, and patient and process outcomes. We also discuss current concepts in this field and highlight

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Methodology & Clinical Focus

Investigation into artificial intelligence models, machine learning algorithms, and clinical predictive tools applied to orthopedic surgery and musculoskeletal medicine.

Key Findings & Analysis

A shared decision-making tool using patient-reported outcomes and clinical data can personalize total joint replacement risks and benefits, improving decision quality and patient satisfaction.

Extended Repository Abstract

Up to one-third of total joint replacement (TJR) procedures may be performed inappropriately in a subset of patients who remain dissatisfied with their outcomes, stressing the importance of shared decision-making. Patient-reported outcome measures capture physical, emotional, and social aspects of health and wellbeing from the patient's perspective. Powerful computer systems capable of performing highly sophisticated analysis using different types of data, including patient-derived data, such as patient-reported outcomes, may eliminate guess work, generating impactful metrics to better inform the decision-making process. We have created a shared decision-making tool which generates personalized predictions of risks and benefits from TJR based on patient-reported outcomes as well as clinical and demographic data. We present the protocol for a randomized controlled trial designed to assess the impact of this tool on decision quality, level of shared decision-making, and patient and process outcomes. We also discuss current concepts in this field and highlight opportunities leveraging patient-reported data and artificial intelligence for decision support across the care continuum.

Clinical Decision Support & Synthesis

Artificial intelligence and machine learning applications in orthopedic surgery demonstrate significant potential for standardizing clinical pathways, augmenting surgical decision-making, and personalizing post-operative rehabilitation. Continuous validation against prospective clinical datasets and multidisciplinary oversight remain paramount for ethical and safe clinical deployment.

Bibliographic Details & Indexing

- **Authors:** P. Jayakumar, K. Bozic
- **Journal / Venue:** Journal of Orthopaedic Research® (2020.0)
- **SJR Metric:** Quartile 1.0
- **Citation Count:** 53 citations recorded
- **Digital Object Identifier (DOI):** https://doi.org/10.1002/jor.24614
- **Consensus Evidence Link:** https://consensus.app/papers/advanced-decision%E2%80%90making-using-patient%E2%80%90reported-jayakumar-bozic/2fb746e1d02b5d208e1d3e2598cf5aea/